



Republic of Iraq
Middle Technical University
Technical Engineering College of Artificial Intelligence
Cybersecurity Engineering Technology Department

Computer Network Protocols

Chapter One

Network Overview

Prepared By:
Assistant Prof. Dr. Mahmoud Shuker Mahmoud
mahmoud.shukur@mtu.edu.iq

1.1- Network Architecture

A set of layers and protocols is called the *network architecture*.

Protocol Hierarchies

- Networks are organized as layers to reduce design complexity.
- Each layer offers services to the higher layers. Between adjacent layers is an interface.

1.2- Network Protocols

Protocol:

A network protocol defines rules and conventions for communication between network devices. Network protocols include mechanisms for devices to identify and make connections with each other, as well as formatting rules that specify how data is packaged into sent and received messages.

Protocol Process:

1. The format or structure of the message
2. The process by which networking devices share information about pathways with other networks
3. How and when error and system messages are passed between devices
4. The setup and termination of data transfer sessions

1.3- Layering in Networked Computing

- OSI Model (Open System Interconnection)
- TCP/IP Model (Transmission Control Protocol/ Internet Protocol)

Why a layered model?

1. Breaks down communication into smaller, simpler parts.
2. Easier to teach communication process.
3. Allows different hardware and software to work together.
4. Reduces complexity.

1.4- The OSI model

The Open Systems Interconnection is the model developed by the International Standards Organization (ISO). The OSI model benefits are:

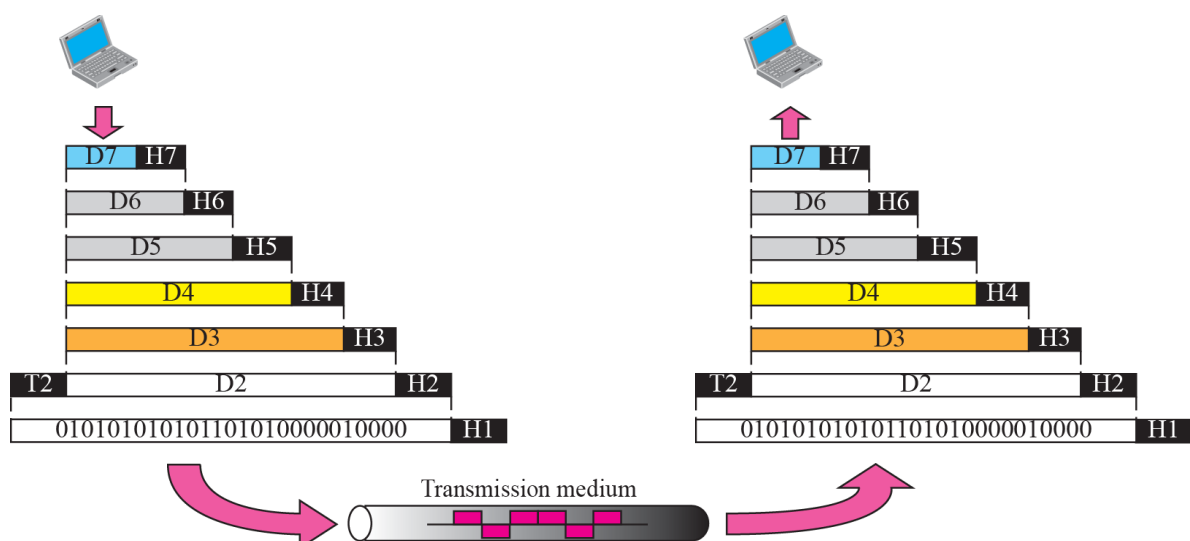
- Helps us understand how data gets from one user's computer to another.
- It aids to provide an organized structure for hardware and software developers to follow.

Why use a reference model?

- Serves as an outline of rules for how protocols can be used to allow communication between computers.
- Each layer has its own function and provides support to other layers.

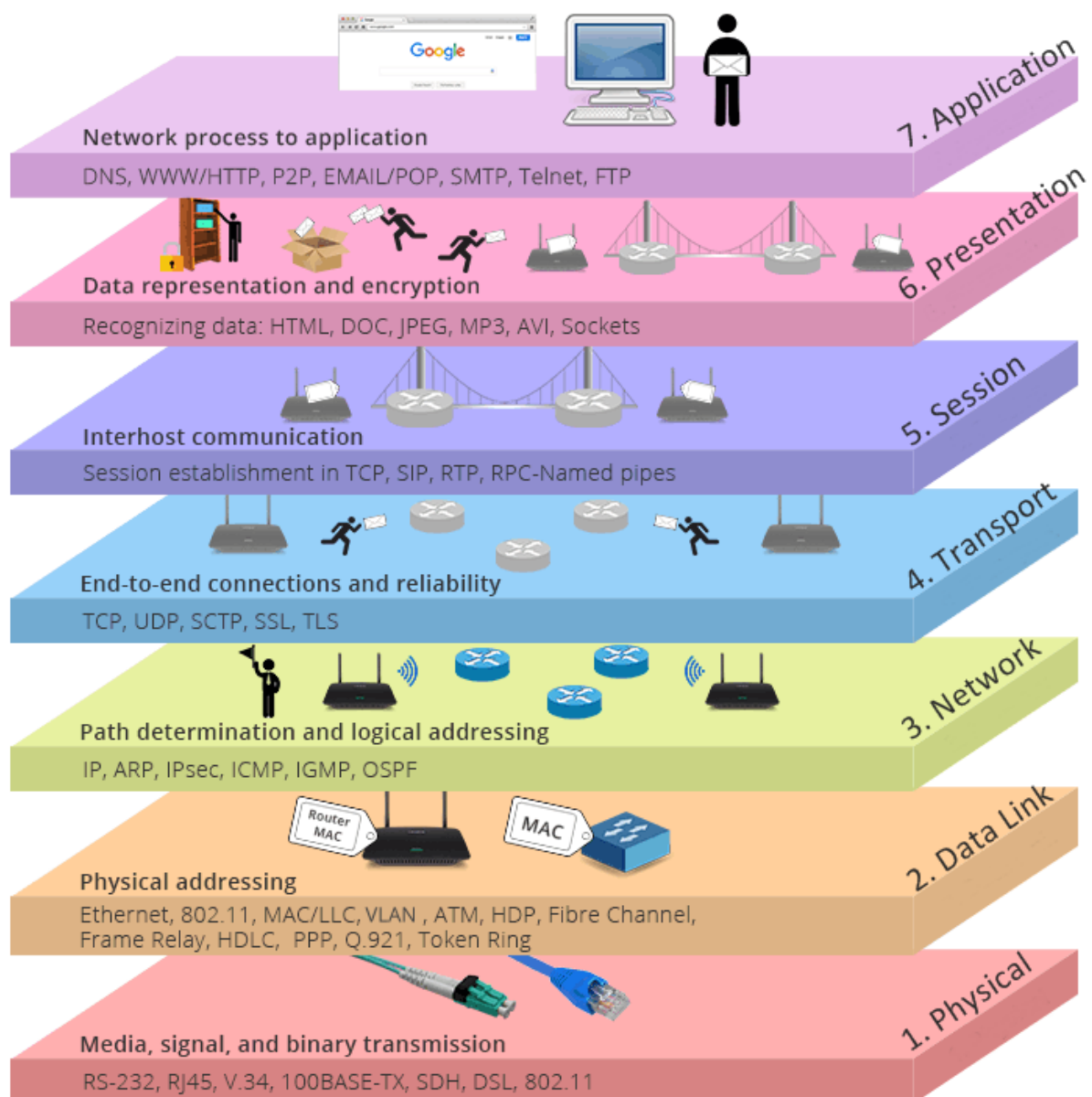
OSI model		
Layer	Name	Example protocols
7	Application Layer	HTTP, FTP, DNS, SNMP, Telnet
6	Presentation Layer	SSL, TLS
5	Session Layer	NetBIOS, PPTP
4	Transport Layer	TCP, UDP
3	Network Layer	IP, ARP, ICMP, IPsec
2	Data Link Layer	PPP, ATM, Ethernet
1	Physical Layer	Ethernet, USB, Bluetooth, IEEE802.11

At each layer, a header (H) can be added to the data unit. At layer 2, a trailer (T) may also be added.



Layer	Main Topics
Physical Layer	• Transmission mediums (transmit bits over medium) • Encoding • Modulation • Multiplexing • Topology • Provide mechanical and electrical specification • Repeaters & Hubs (multi-port repeater)
Data Link Layer	• Node to node (hop to hop) delivery • Frame format • Physical addressing (MAC Address) • Access control • Flow control • Error handling • Bridges & Switches (multi-port bridges)
Network Layer	• Internetworking • Controls the operation of the subnet. • Routing algorithms (Routing packets from source to destination) • Internet Protocol (IP) addressing (Logical addressing) • Routers
Transport Layer	• Connection-oriented and connectionless services • Provide reliable process to process message delivery & error recovery • Transmission Control Protocol (TCP) • User Datagram Protocol (UDP) • Provides additional Quality of Service. • Port address (service point address) • End-to-end flow control. • Segmentation and reassembly.
Session Layer	• Allows users on different machines to establish sessions (dialogue) between them. • managing dialogue control. • Session management. • Synchronization. • Graceful close

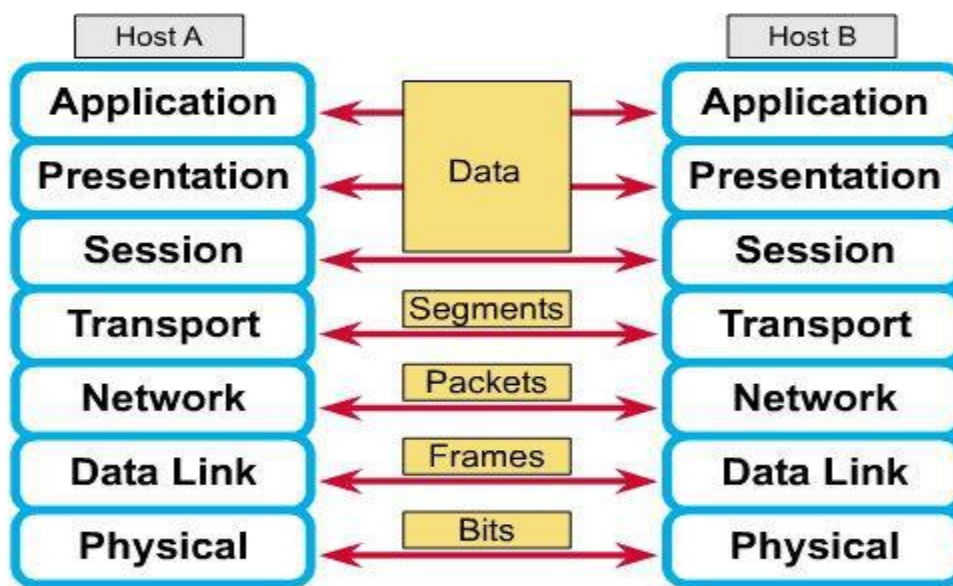
Presentation Layer	• Concerned with the syntax and semantics of the information. • Data Translation to preserve its meaning. • Data compression. • Data encryption. • Data Security
Application Layer	• Provides protocols that are commonly needed. • To allow access to network resource • (FTP), (HTTP), (SMTP), (SNMP),(NFS),(Telnet)



1.5- Data Encapsulation

Each layer contains a Protocol Data Unit (PDU)

1. PDU's are used for peer-to-peer contact between corresponding layers.
2. Data is handled by the top three layers, then Segmented by the Transport layer.
3. The Network layer places segment into packets and the Data Link frames the packets for transmission.
4. Physical layer converts frame to bits and sends it out over the media.
5. The receiving computer reverses the process using the information contained in the PDU (headers of each layer).

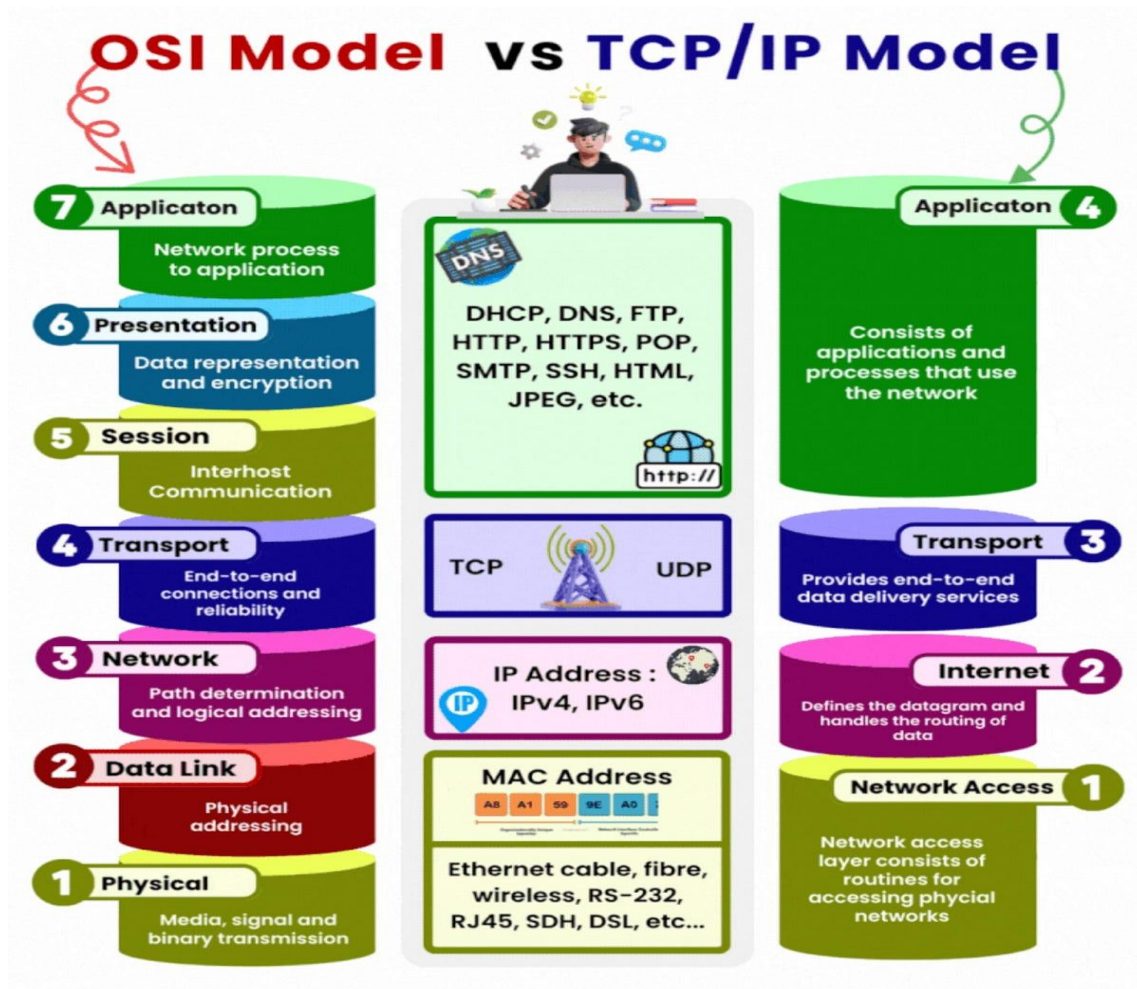


1.6- TCP/IP Model

The **TCP/IP protocol suite** was developed prior to the OSI model. Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers:

- Layer 4: Application
- Layer 3: Transport
- Layer 2: Internet
- Layer 1: Network access

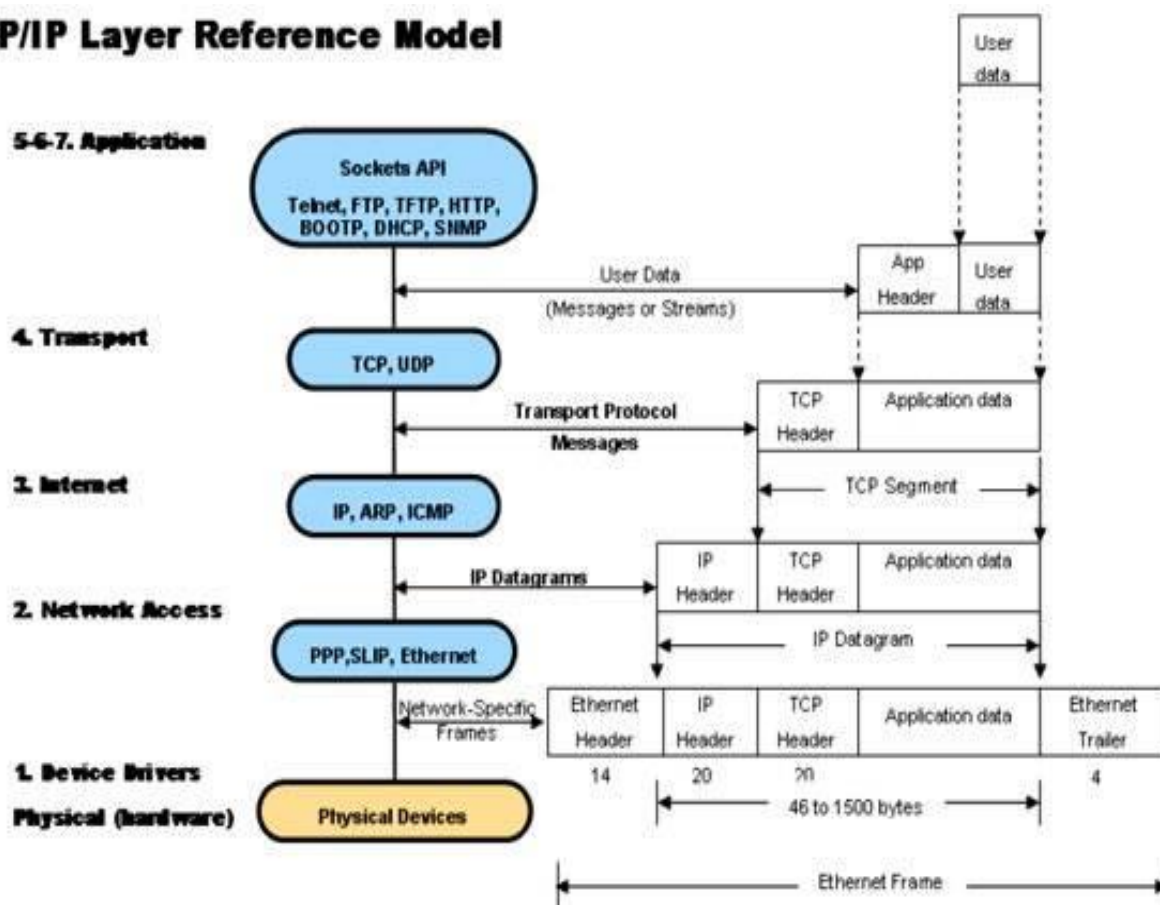
It is important to note that some of the layers in the TCP/IP model have the same name as layers in the OSI model. Do not confuse the layers of the two models.



Q: Compare between OSI and TCP/IP reference Models

Parameters	OSI Model	TCP/IP Model
No. of Layers	7 layers	4 Layers
Acronym	Open System Interconnection	Transmission Control Protocol/ Internet Protocol
Developed by	ISO (International Standard Organization)	DoD (Department of Defence)
Layer Separation	Has a separate session and presentation layers	Combine the session and presentation layers into the application layer
Protocol Implementation	Model was developed before the development of protocols	Protocols were developed then the model is developed
Model implementation	It is a theoretical model	It is the model around it the internet is developed
Reliable Delivery	It gives guarantee reliable packet delivery because it supports both connection oriented and connectionless communication in network layer	It does not give guarantee reliable packet delivery because it supports only connectionless communication in network layer

The TCP/IP Layer Reference Model



Q /Explain the delivery of data in Layered model?

Type of delivery	Layer	Shape of data	Type of addressing
End to End	Transport	Segment	Port (socket)
Source To Destination	Network	Packet	Logical (IP)
Node to Node	Data Link	Frame	Physical (MAC)

1.7- Categories of Networks

The category of a network can be determined according to its size into:

- *Local area networks (LANs).*
- *Metropolitan area networks (MANs).*
- *Wide area networks (WANs).*

1.7.1- Local Area Network (LAN)

A local area network is usually privately owned and links the devices in a single office building, or campus. A LAN can be as simple as two PCs and a printer in someone's home office or it can extend throughout a company and include voice, sound, and video peripherals.

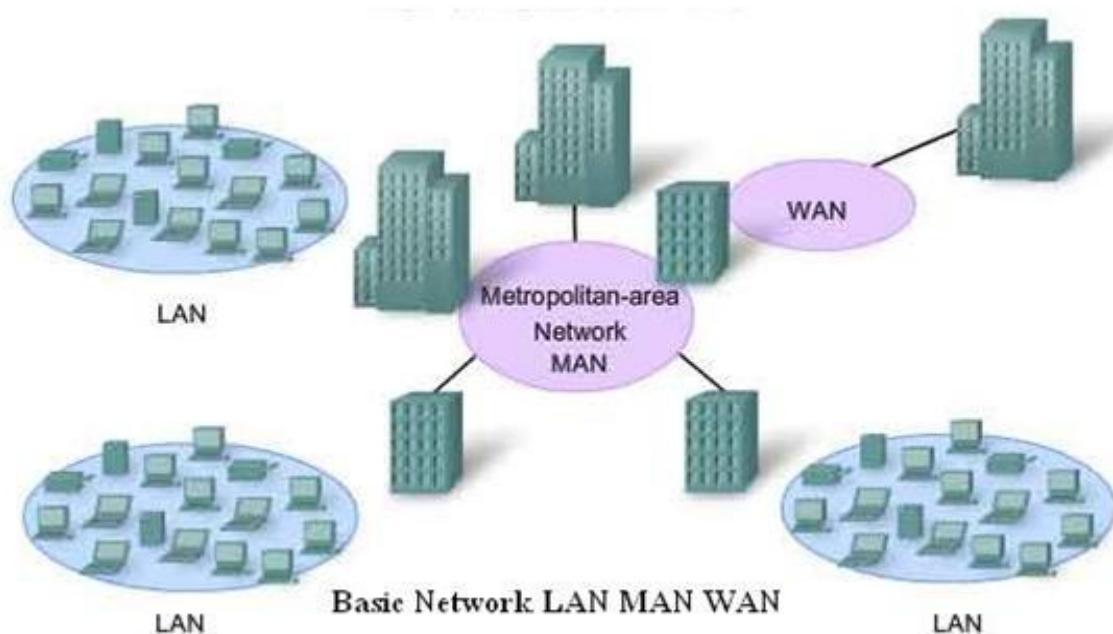
1.7.2- Metropolitan Area Network (MAN)

A metropolitan area network is designed to *extend over* an entire city. It may be a single network such as a cable television network or it may be a means of connecting a number of LANs into a larger network so that resources may be shared LAN-to-LAN as well as device-to-device. For example, a company can use a MAN to connect the LANs in all of its offices throughout a city.

A MAN may be wholly owned and operated by a private company, or it may be a service provided by a public company.

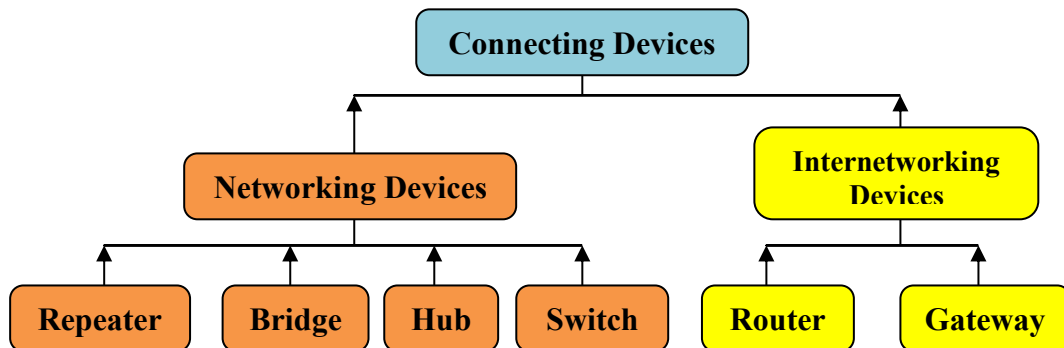
1.7.3- Wide Area Network (WAN)

A wide area network provides *long-distance* transmission of data; voice, image, and video information over large geographical areas that may comprise a country, a continent, or even the whole world. In contrast to LANs (which depend on their own hardware for transmission), WANs may utilize public, leased, or private communication devices, usually in combinations, and can therefore span an unlimited number of miles. A WAN that is wholly owned and used by a single company is often referred to as an *enterprise network*.

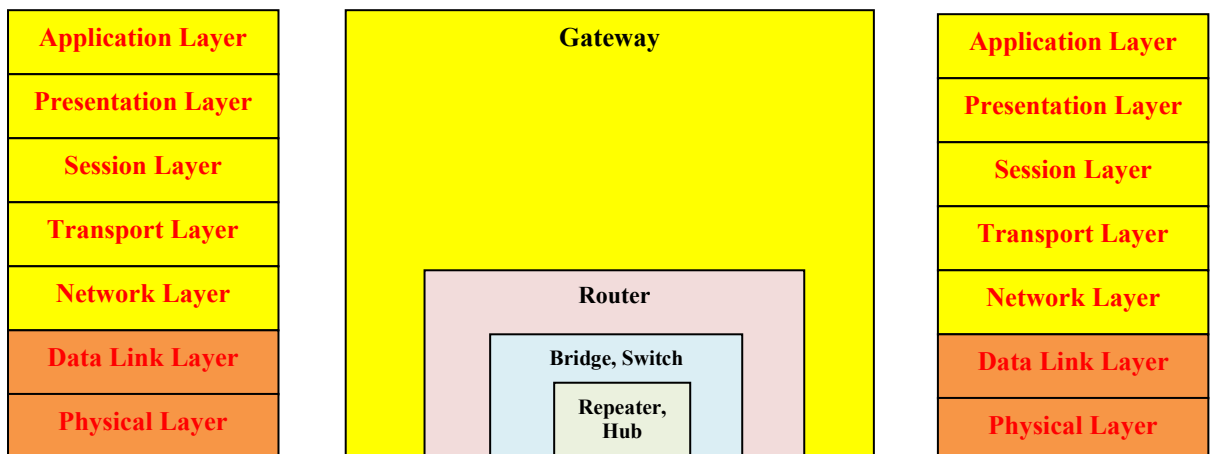


1.8- Networking and Internetworks

To connect a network or multiple networks together there are different types of connecting devices as illustrated in figure below:

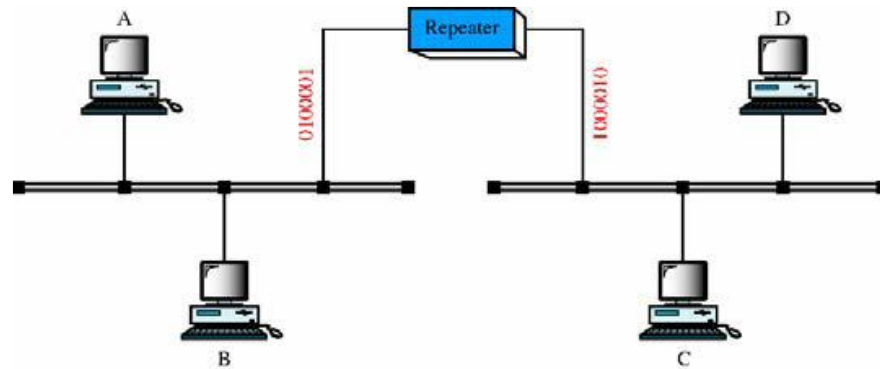


Each of these devices operates at different layer of the OSI model as shown in figure below:



1.8.1- Repeater:

Devices used to extend the network cable length beyond the limit of the specified cable. The purpose of a repeater is to **regenerate and retime network signals** at the bit level to allow them to travel a longer distance on the media. Repeaters are classified as **Layer 1 devices** in the OSI model, because they act only on the bit level and look at no other information.

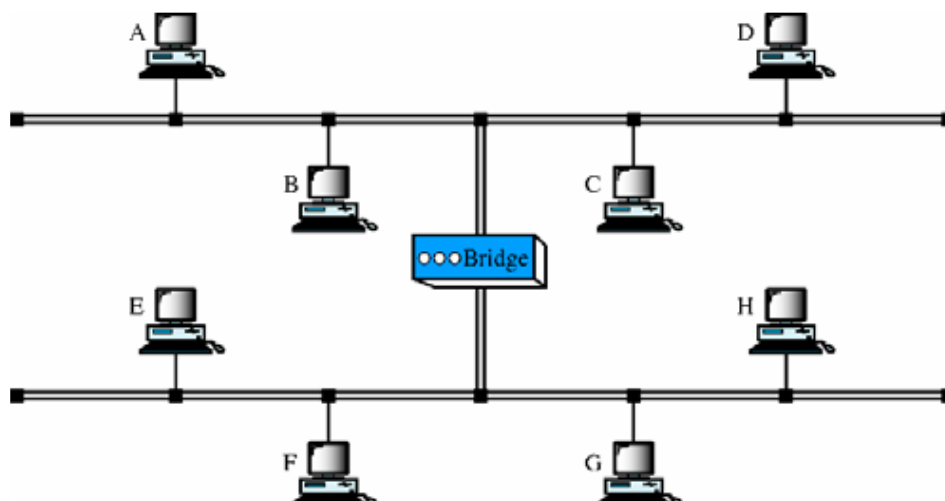


1.8.2- Hubs

The purpose of a hub is to **regenerate and retime network signals**. This is done at the bit level to a large number of hosts (e.g. 4, 8, or even 24) using a process known as concentration. You will notice that this definition is very similar to the repeaters, which is why a hub is also known as *a multi-port repeater*. Hubs are considered **Layer 1 devices** because they only regenerate the signal and broadcast it out all of their ports (network connections).

1.8.3- Bridges

A bridge is a **Layer 2 device** designed to connect two LAN segments. The purpose of a bridge is to filter traffic on a LAN, to keep local traffic local, yet allow connectivity to other parts (segments) of the LAN for traffic that has been directed there.



1.8.4 Switches

A switch is a **Layer 2 device** just as a bridge is. In fact, a switch is called a **multi-port bridge**, just like a hub is called a multi-port repeater. The difference between the hub and switch is that switches make decisions based on MAC addresses and hubs don't make decisions at all. Because of the decisions that switch make, they make a LAN much more efficient. They do this by "switching" data only out the port to which the proper host is connected. In contrast, a hub will send the data out all of its ports so that all of the hosts have to see and process (accept or reject) all of the data.

1.8.5- Routers

A router is a **Layer 3 device**. Working at Layer 3 allows the router to make decisions based on groups of network addresses as opposed to individual Layer 2 MAC addresses. Routers can also connect different Layer 2 technologies, such as Ethernet, Token-ring, and FDDI. However, because of their ability to route packets based on Layer 3 information, routers have become the backbone of the Internet, running the IP protocol. The purpose of a router is to examine incoming packets (Layer 3 data), choose the best path for them through the network, and then switch them to the proper outgoing port. A router is a type of internetworking device that passes data packets between networks, based on Layer 3 addresses. A router has the ability to make intelligent decisions regarding the best path for delivery of data on the network.

1.8.6- Gateway

A gateway is a **layer 7 device** and can be hardware or software, it works as a network connectivity device that connects two different configuration networks. Gateways are also known as protocol converters, because they play an important role in converting protocols supported by traffic on different networks. As a result, it allows smooth communication between two networks. It works as the entry-exit point for a network because all traffic that passes across the networks must pass through the gateway. A gateway monitors and controls all the incoming and outgoing network traffic.